

635 Homework 9 answers

1. (a) $\psi^{TE_1} = \sin(k_x x) \cos(k_y y) e^{\pm j k_z z}$

$$k_x = \frac{m\pi}{a}, m=1,2,3,\dots; k_y = \frac{n\pi}{b}, n=0,1,2,3,\dots$$

$$E_x = 0 \quad k_z = \sqrt{k^2 - k_x^2 - k_y^2}$$

$$E_y = \frac{\pm j k_z}{\epsilon} E_0 \sin(k_x x) \cos(k_y y) e^{\pm j k_z z}$$

$$E_z = -\frac{k_y}{\epsilon} E_0 \sin(k_x x) \sin(k_y y) e^{\pm j k_z z}$$

(b) $B = \frac{k_y \omega \epsilon}{k_x k_z}$

$$A = \frac{-j k_x k_z}{(k^2 - k_z^2)}$$

2. $\psi_{mn} = \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} \frac{-j\epsilon}{k_{z,mn}} \frac{\epsilon_m}{ab} \cos(\frac{m\pi}{a} x_s) \cos(\frac{n\pi}{a} x) \sin(\frac{n\pi}{b} y) \int_{y'=0}^b I_m(y') \sin(\frac{n\pi}{b} y') dy'$
or y_s
 $x e^{\pm j k_{z,mn} z}$

$$E_x = \frac{1}{\epsilon} \frac{\partial}{\partial z} \psi_{mn}$$

$$= -\sum_{m=0}^{\infty} \sum_{n=1}^{\infty} \left(\frac{\epsilon_m}{ab}\right) \cos(\frac{m\pi}{a} x_s) \cos(\frac{n\pi}{a} x) \sin(\frac{n\pi}{b} y) \int_{y'=0}^b I_m(y') \sin(\frac{n\pi}{b} y') dy'$$

 $x e^{-j k_{z,mn} z}$
 $z > 0$

$$\epsilon_m = \begin{cases} 1; & m=0 \\ 2; & m>0 \end{cases}$$